

REQUIREMENT ANALYSIS AND LITERATURE SURVEY

Phishing URL Detection

2.1 Requirement Analysis

2.1.1 Introduction

Phishing is one of the fastest-growing cyber threats, where attackers use fraudulent websites and URLs to trick users into revealing confidential information such as usernames, passwords, banking details, and personal data. Traditional phishing detection methods mainly rely on blacklists and manually maintained databases, which are ineffective against newly created phishing websites. This project proposes a machine learning-based web application capable of detecting phishing URLs in real time by analyzing URL characteristics and predicting whether a URL is legitimate or malicious.

2.1.2 Problem Statement

Cybercriminals continuously generate new phishing websites that closely resemble legitimate websites. Existing blacklist-based detection systems fail to identify these newly created (zero-day) phishing URLs because they are not yet included in their databases. This creates a need for an intelligent detection system that can identify phishing URLs based on their characteristics rather than relying solely on previously known malicious URLs.

2.1.3 Objectives

The major objectives of the proposed system are:

- To develop an intelligent phishing URL detection system using machine learning.
- To classify URLs as legitimate or phishing with high accuracy.
- To build a user-friendly web application for real-time URL verification.
- To reduce users' risk of phishing attacks by providing instant predictions.
- To create a scalable system that can be retrained with new datasets as phishing techniques evolve.

2.1.4 Functional Requirements

The proposed system should be capable of:

- Accepting a URL entered by the user.
- Validating the entered URL.
- Extracting important URL features such as URL length, number of dots, special characters, HTTPS usage, presence of IP addresses, and suspicious keywords.
- Processing the extracted features using a trained machine learning model.
- Predicting whether the URL is phishing or legitimate.
- Displaying the prediction result along with the confidence score.
- Maintaining a history of scanned URLs.
- Providing an easy-to-use graphical web interface.

2.1.5 Non-Functional Requirements

The system should satisfy the following non-functional requirements:

- High prediction accuracy.
- Fast response time (less than a few seconds).
- Secure processing of user inputs.
- User-friendly and responsive interface.
- Reliability and availability.
- Scalability for future model improvements.
- Easy maintenance and software updates.

2.1.6 Hardware Requirements

- Processor: Intel Core i3 or above
- RAM: Minimum 4 GB (8 GB recommended)
- Storage: 20 GB available disk space
- Internet Connection

2.1.7 Software Requirements

- Operating System: Windows/Linux/macOS
- Programming Language: Python
- IDE: Visual Studio Code or PyCharm
- Backend Framework: Flask or Django
- Frontend Technologies: HTML, CSS, JavaScript
- Machine Learning Library: Scikit-learn
- Data Processing Libraries: Pandas and NumPy
- Visualization Library: Matplotlib
- Database: SQLite or MySQL (optional)
- Browser: Google Chrome, Mozilla Firefox, Microsoft Edge

2.2 Literature Survey

2.2.1 Introduction

Phishing attacks continue to be among the most common forms of cybercrime. Researchers have developed several approaches for phishing detection, including blacklist-based methods, heuristic approaches, machine learning algorithms, and deep learning techniques. Machine learning has emerged as one of the most effective approaches because it can identify previously unseen phishing websites by learning patterns from datasets.

2.2.2 Review of Existing Studies

Study 1: Machine Learning-Based Phishing Website Detection

Researchers compared algorithms such as Decision Tree, Random Forest, Support Vector Machine (SVM), Logistic Regression, and Naïve Bayes for phishing URL detection. The study concluded that Random Forest achieved the highest accuracy due to its ensemble learning capability and resistance to overfitting.

Study 2: URL Feature-Based Phishing Detection

This study focused on lexical features extracted directly from URLs, including URL length, number of dots, hyphens, special characters, presence of IP addresses, HTTPS protocol usage, and suspicious keywords. The results demonstrated that URL-based analysis enables fast and efficient phishing detection without downloading webpage content.

Study 3: Deep Learning for Phishing Detection

Recent research has explored deep learning models such as Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) for phishing detection. These models achieved high accuracy but required larger datasets and greater computational resources than traditional machine learning methods.

Study 4: Ensemble Learning Approaches

Several researchers proposed ensemble learning techniques such as Random Forest, XGBoost, and AdaBoost for phishing detection. These approaches combine multiple decision trees, improving prediction accuracy and reducing overfitting.

2.2.3 Existing System

The existing phishing detection systems mainly rely on:

- Blacklist databases
- Browser-based warning systems
- Rule-based detection methods

Limitations

- Unable to detect zero-day phishing websites.
- Requires continuous updates to blacklist databases.
- Lower accuracy for newly generated phishing URLs.
- Limited adaptability to evolving phishing techniques.

2.2.4 Proposed System

The proposed system uses machine learning algorithms to classify URLs based on their extracted features instead of relying only on blacklist databases. The system analyzes lexical characteristics of URLs, processes them through a trained machine learning model, and predicts whether the URL is legitimate or phishing. The result is presented through a web-based interface, allowing users to verify suspicious URLs quickly and efficiently.

2.2.5 Research Gap

Although existing phishing detection systems perform well against known phishing websites, they struggle to identify newly created phishing URLs. Most traditional systems depend on blacklists or manually defined rules that require frequent updates. There is a need for an intelligent and adaptive detection system capable of learning phishing patterns from large datasets and accurately identifying unknown phishing websites in real time.

2.2.6 Conclusion

The literature survey indicates that machine learning is an effective approach for phishing URL detection. Algorithms such as Random Forest, XGBoost, Support Vector Machine, and Decision Tree have demonstrated high accuracy in detecting phishing URLs based on lexical features. Integrating these algorithms into a web application provides users with a fast, reliable, and scalable solution for real-time phishing detection. The proposed project addresses the limitations of traditional blacklist-based methods by employing machine learning to improve the detection of both known and previously unseen phishing URLs.